Main idea

How many physical polarizations are there?

$A_\mu = \epsilon_\mu e^{ikx} + c.c.$ plug in $\partial_\mu A^\mu = 0 \Rightarrow \boxed{k^\mu \epsilon_\mu = 0}$ $\boxed{\epsilon_\mu \to \epsilon_\mu - \lambda k_\mu}$

$\underbrace{\text{Mathematical possibilities}}$ $-$ $\underbrace{\text{constraints}}_{1}$ $-$ $\underbrace{\text{equivalence}}_{1}$

μ has 4 possibilities

$= \underbrace{\text{physical polarizations}}$

2 polarizations.

$\Downarrow$

analogy: gravitational waves?

polarizations.

$\epsilon_{\mu\nu} \to \epsilon_{\mu\nu} - \lambda_\mu k_\nu - \lambda_\nu k_\mu$

10 ϵ's $-$ 4 constraints $-$ 4 equivalences (residual gauge freedom).

indep. in $d=4$ for $h_{\mu\nu}$

$\boxed{k^\mu \epsilon_{\mu\nu} - \frac{1}{2} k_\nu \epsilon^\alpha_\alpha = 0}$ $= \boxed{\text{2 physical polarizations}}$ as well.

What happens to an object at rest when a gravitational wave pass through?

$\tau = t$ when particle at rest

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} = 0$$

$\Gamma^\mu_{00} = \frac{1}{2} g^{\mu c} (g_{0c,0} + g_{c0,0} - g_{00,c}) = 0$

convince myself that this is valid.

the object

$\boxed{h_{\mu\nu} = \epsilon_{\mu\nu} e^{ikx} + c.c.}$

or $\epsilon_{\mu\nu}$

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\boxed{g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}}$

the particle doesn't seem to do anything in the presence of the gravitational wave, but that is an artifact of the coord we choose $\Rightarrow$ spacetime itself does get stretched!

we'll see.

$$\frac{d^2\Delta^\mu}{d\tau^2} = R^\mu{}_{\alpha\beta\gamma}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}\Delta^\gamma$$

in weak fields:

$$\frac{d^2\Delta^\mu}{dt^2} = R^\mu{}_{00\gamma}\Delta^\gamma$$

$$R_{\alpha\beta\gamma\delta} = \frac{1}{2}\left(h_{\alpha\delta,\beta\gamma} - \cancel{h_{\alpha\gamma,\beta\delta}} + \cancel{h_{\beta\gamma,\alpha\delta}} - \cancel{h_{\beta\delta,\alpha\gamma}}\right)$$

$$R_{\alpha 00\delta} \Rightarrow \frac{1}{2}\left(h_{\alpha\delta,00}\right)$$

$$\boxed{= \frac{1}{2}\frac{d^2}{dt^2}h_{\alpha\delta}}$$

$\boxed{\alpha,\beta = (x,y)}$

because

$$\epsilon_{\mu\nu} = \begin{pmatrix} & & & \\ & & & \\ & & h_+ & h_\times \\ & & h_\times & -h_+ \end{pmatrix}$$

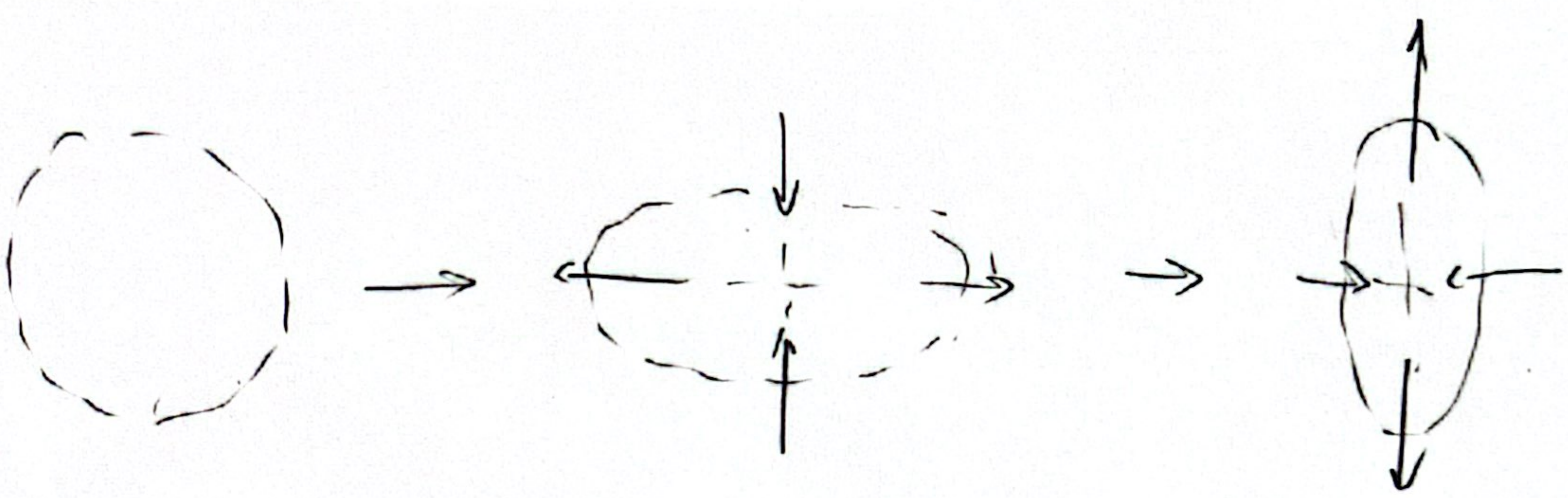
Choose $h_\times = 0$; $h_+ \neq 0$

$\Rightarrow h_{xx} = h_+\cos(\omega t)$ $\quad \frac{d^2}{dt^2}h_{xx} = -\omega^2 h_+\cos(\omega t)$

$h_{yy} = -h_+\cos(\omega t)$ $\quad \frac{d^2}{dt^2}h_{yy} =$

$$\frac{d^2\Delta^x}{dt^2} \propto -\omega^2 h_+\cos(\omega t)\,\Delta^x$$

$$\frac{d^2\Delta^y}{dt^2} \propto +\omega^2 h_+\cos(\omega t)\,\Delta^y$$

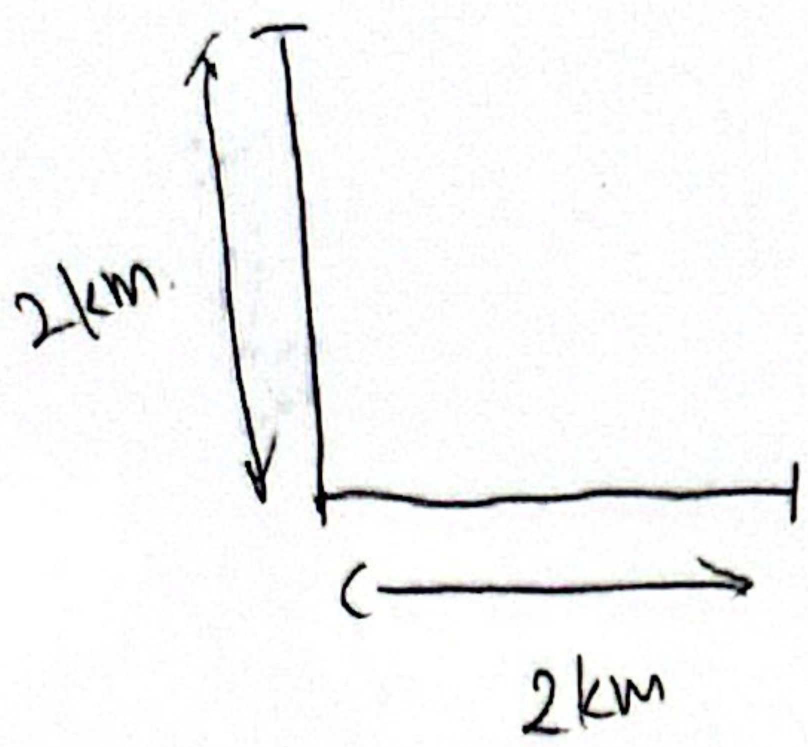


→ relates to LIGO

+ dir

coordinates didn't change,
but the metric changes.

How are waves generated?



Ligo can measure $h > 10^{-22}$

$\Delta \text{length} = 10^{-21} \cdot 10^{3} m = 10^{-18} m$

Bohr radius of Hydrogen atom: $10^{-10} m$.

Nucleus: $10^{-15} m$.

$$h_{ij} \sim \frac{2 G_N \frac{d^2}{dt^2}\left(I_{ij} - \frac{1}{3}\delta_{ij} I^{\alpha}{}_{\alpha}\right)}{c^4 (\text{distance to the object})}$$

dimensionless

2 solar mass object: 2×10^{30} kg

10 My away

2 obj 10 km away: $h \sim 10^{22}$

VIRGO

(3 locations: point to where the wave came from).

36×10^2
$\times 25$
100

$(900 \times 10^2) \times 365$

3.65
$\times 0.9$
3.3

★ Green's Function

★ how to use dimensional analysis to guess the formula of h_{ij} → very cool tool!